

AI Smart Link Manager - Detailed Concepts Guide

A comprehensive revision guide covering concepts learned during Day 3.

Frontend Concepts

React state management, useState, controlled inputs, conditional rendering, loading indicators, error states, fetch API, JSON parsing, and dynamic UI updates.

Backend Concepts

Express server creation, routes, middleware, request objects, response objects, JSON communication, asynchronous functions, async-await, and exception handling.

API Concepts

API keys, authentication, quotas, rate limits, SDK usage, request-response lifecycle, prompt design, response parsing, and structured outputs.

Database Concepts

Databases, tables, records, columns, primary keys, auto increment, SQL queries, SELECT, INSERT, connection objects, query execution, and persistence.

Caching Concepts

Database caching, duplicate detection, reducing expensive API calls, improving efficiency, and reusing previously generated results.

Git and GitHub Concepts

Repository creation, git add, git commit, git push, version control, remote repositories, and project documentation.

Environment Variables

Purpose of .env files, secure configuration management, hiding credentials, and preventing sensitive information from entering source control.

Interview Preparation Notes

Be prepared to explain architecture, Gemini integration, MySQL schema, duplicate URL detection logic, API flow, database usage, and reasons for caching.

Key Takeaway

The most important learning was understanding how frontend, backend, AI services, databases, and version control systems work together in a real-world application.